

Does genetic-fuzzy control improve autonomous vehicle path tracking?

Yes, **genetic-fuzzy control generally improves path tracking** accuracy and robustness compared with non-optimized or conventional controllers.

Does genetic-fuzzy control improve autonomous vehicle path tracking? N = 9

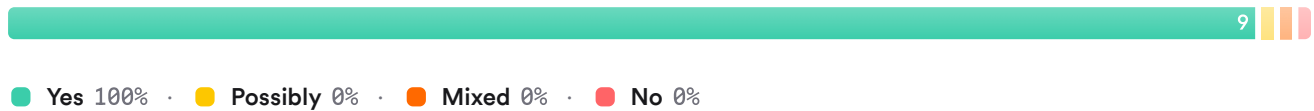


FIGURE 1 Consensus on GA-optimized fuzzy control for tracking

Genetic-fuzzy control combines fuzzy logic controllers (FLCs) with **genetic algorithms (GAs)** that automatically tune fuzzy rules, gains, or membership functions. Across ground vehicles, AGVs, bicycles, robots, and UAVs, GA-tuned fuzzy or hybrid fuzzy controllers consistently show better tracking than untuned fuzzy or more conventional controllers.

Effectiveness and Comparisons

- **Road and delivery vehicles:**
 - A multi-fuzzy genetic tracking controller for an autonomous delivery van reduces RMS tracking error by ~66–91% versus manually or geometrically weighted fuzzy controllers (■ Ghazali et al., 2025).
 - A GA-optimized fuzzy steering controller for unmanned vehicles robustly tracks reference trajectories with high accuracy in simulations and experiments (■ Guo et al., 2012).
 - MPC-GA path tracking for an autonomous car achieves efficient tracking in simulation and on-road tests and outperforms “traditional methods” (not GA-based) (Wang et al., 2022).
- **AGVs and industrial vehicles:**
 - GA-optimized fuzzy trajectory tracking for a hybrid AGV improves guiding error while also optimizing speed and energy use, beating an optimized PID controller on four paths (Sierra-García & Santos, 2024).
 - GA-selected fuzzy membership functions enable accurate path estimation and tracking of automated vehicles in complex environments (■ Rajurkar et al., 2016).
- **Robots, bicycles, UAVs:**
 - GA-tuned fuzzy controllers for an autonomous bicycle improve path tracking and balance in simulations (■ Chen & Dao, 2007).
 - Metaheuristic-optimized fuzzy controllers (including GA) for robot path tracking achieve lower RMSE than previously published controllers (Mancilla et al., 2022).
 - GA-tuned fuzzy gains for a UAV allow completion of trajectories that an untuned fuzzy controller cannot follow and cut tracking errors by up to ~30% in some tasks (■ Rodríguez-Abreo et al., 2024).
 - GA-tuned fuzzy/PID-type-2 controllers for quadrotor trajectory tracking show improved robustness and numerical performance indices versus backstepping and type-1 fuzzy controllers (Moali et al., 2024).

Example Outcomes Across Platforms

Platform / vehicle type	GA-fuzzy impact on tracking	Citations
Delivery van (road)	Up to ~90% lower RMS tracking error	(■ Ghazali et al., 2025)
Unmanned road vehicle	High-accuracy tracking in varied conditions	(■ Guo et al., 2012 ; Wang et al., 2022)
Industrial AGV	Lower guiding error than optimized PID	(Sierra-García & Santos, 2024 ; ■ Rajurkar et al., 2016)
Bicycle / mobile robots	Effective simulated tracking with lower RMSE	(■ Chen & Dao, 2007 ; Mancilla et al., 2022 ; ■ Tolossa et al., 2024)
UAV / quadrotor	Enables harder paths; lower errors, robust	(■ Rodríguez-Abreo et al., 2024 ; Moali et al., 2024)

FIGURE 2 Reported benefits across different vehicle types

Mechanisms and Practical Considerations

- GAs tune **membership functions, gains, and controller weights**, avoiding manual rule tuning and improving robustness under nonlinear dynamics and disturbances ([Sierra-García & Santos, 2024](#); [Moali et al., 2024](#); [■ Chen & Dao, 2007](#); [■ Rodríguez-Abreo et al., 2024](#); [■ Guo et al., 2012](#)).
- Benefits are shown mainly in **simulation and controlled experiments**; performance depends on chosen cost function (e.g., RMS error, energy, safety), so “improvement” is metric-specific ([Sierra-García & Santos, 2024](#); [Mancilla et al., 2022](#); [Wang et al., 2022](#); [■ Al-Bayati et al., 2025](#)).
- Some studies note risks of **overtraining** if optimization is done on a single path or metric, highlighting the need to validate on multiple trajectories ([Mancilla et al., 2022](#)).

Conclusion

Across many platforms, GA-optimized fuzzy (genetic-fuzzy) controllers reliably improve path or trajectory tracking versus untuned fuzzy, PID, or other baseline methods, and often add robustness to disturbances. Gains are demonstrated mainly in simulation and targeted experiments, and depend on good choice of objective functions and validation scenarios.

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